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Toward Real-Time Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

Shunyang Li | Benchmarks measured | 27 Oct 2026



Agenda

01 Toward Real-Time H3 Generation

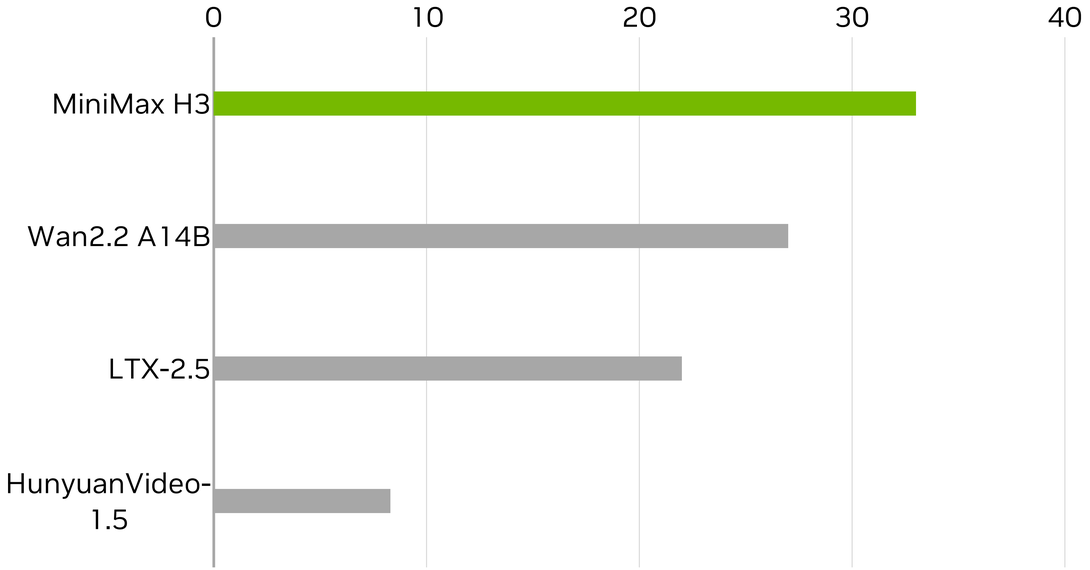
02 vLLM-Omni and ComfyUI Applications



01 · Real-Time H3 Generation

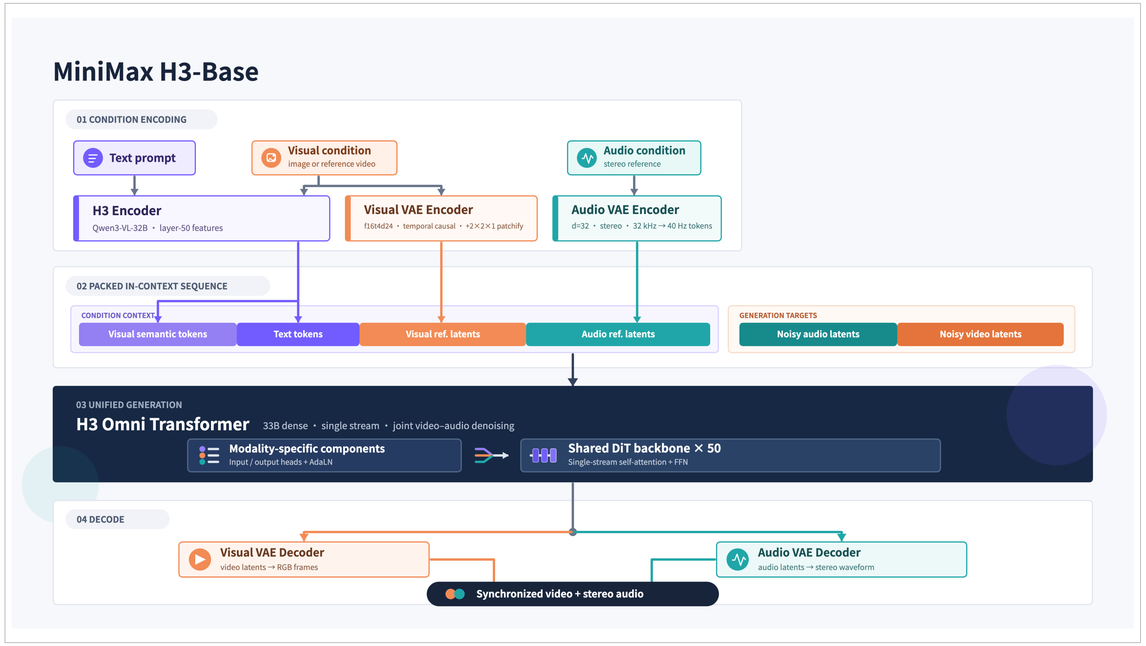
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

A 32B encoder feeds the 33B Omni Transformer and visual/audio VAEs. Fifty blocks × 50 steps dominate compute; about 13B AdaLN parameters can be cached

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

SM120 Brings FP4 Compute; PCIe Limits Scale

All discrete SM120 cards pair fifth-generation Tensor Cores with GDDR7 and PCIe Gen5; none provides GPU-to-GPU NVLink

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 ×8 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: RDMA via NIC*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

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Baseline: Configuration and Breakdown

Base50 configuration · fresh HTTP timing + separately recorded historical Nsight profile

OUTPUT	VIDEO PER REQUEST	PARALLELISM	HARDWARE
960 × 544	5.167 seconds	8 GPUs	SM120 / PCIe
Reference image: 1344 × 768	124 frames at 24 FPS · seed 43	TP1 · Ulysses 8 · Encoder TP8 · VAE 8	PRO 5000 reference node · local NCCL

251.303 s

Fresh unprofiled HTTP median · n = 3

252.773 s

Separate warmed Nsight request

Profile: mean GPU operation time*

NCCL communication	123.0 s	49.2%
DiT attention + GEMM / MLP	84.5 s	33.8%
AdaLN staging + device layout	38.5 s	15.4%
Encoder, VAE + other GPU work	4.0 s	1.6%

Total: 250.0 s per GPU · 100%

* Kernel + memcpy + memset time, summed across GPUs ÷ 8.
These are exclusive GPU cost buckets, not E2E stage latencies.
NCCL time includes synchronization and peer waiting.

Baseline configuration

Model / steps
Base Ref2VA · BF16 DiT · 50 evaluations · no adapter

Attention / execution
Dense cuDNN attention · token refiner: Torch SDPA
Regional torch.compile · no head-band CUDA Graph

Communication
Local NCCL · NCCL_IB_DISABLE=1 · RDMA off
No A2A permute · no 2 / 3 / 2 head pipeline

Decoder / output
Native unquantized VAE · 8-way spatial tiling
FP16 decode autocast · response encoding after generation

AdaLN
Default CPU weight offload + 256 MiB exact cache
Cache misses stage weights · no offline schedule table

Historical default-offload baseline. An optimized BF16 baseline is pending. Full CPU-stage breakdown and provenance are linked from the title.

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Five-Second Avatar: Live Conversation

Demo focus · reference identity, expression, native speech and continuity

01 · Natural presence

Listen with natural presence.
Microphone stays active; live captions show the question.

02 · Speech and expression

Generate a short answer as native audio and video.
Show generation time and actual reply-video start time.

03 · Continuous generation

Carry video and audio history across windows.
Inspect the face, mouth and transitions while speaking.

960 × 544 · 24 FPS · 8 steps · 8 SM120 GPUs · QAD NVFP4 / native FP4 VSA

119 new frames / 4.958 s per continuation · 20% history noise · video prefix off.
ASR and LLM share GPUs 6 / 7. Full configuration and timing definitions are on slide 21.

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Three Ideas Behind Faster Generation

Technical overview · detailed algorithms and source excerpts remain available on hidden slides

Less model work

Few-step distillation reduces evaluations.
Sparse attention concentrates work on selected regions.
Quantized inference lowers arithmetic and memory cost.

Overlap across eight GPUs

Ulysses distributes attention across heads.
RDMA carries the exchanges.
Head bands overlap transfer with attention compute.

Keep the output path moving

CUDA Graph replays stable head-band compute.
Exact AdaLN caching avoids repeated projections.
NVFP4 VAE and MP4 pre-encoding reduce output cost.

These are optimization families. Use each demo’s own recipe and measurements; the four-step avatar and slide 18 are distinct workloads.

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Cumulative Optimization: Measured Latency

960 × 544 · 124 frames / 24 FPS · 8 exclusive GPUs · same inputs and seed 43

Stage	Cumulative change	Median (s)	Saved (s)	RTF
00	Base50 · original BF16 / dense / default offload	251.303	—	48.639
01	+ PDD · 50 → 8 evaluations	44.327	+206.976	8.579
02	+ Trained VSA · native 8-step model / runtime	25.915	+18.412	5.016
03	+ QAD NVFP4 DiT / native FP4 attention	19.460	+6.455	3.767
04	+ RDMA / direct layout / V-gate overlap	6.832	+12.628	1.322
05	+ 2 / 3 / 2 head pipeline	6.407	+0.425	1.240
06	+ Compute CUDA graphs	6.193	+0.214	1.199
07	+ 36-layer NVFP4 VAE / MP4 pre-encoding	5.420	+0.773	1.049
08	+ Fused tile preparation / early AAC	5.194	+0.226	1.005
09	+ Direct output packing	5.147	+0.047	0.996
10	+ Direct receive-to-tile input	5.060	+0.087	0.979
11	+ Integer routing compaction	5.015	+0.045	0.971
12	**+ Producer Q/K/V copy elision**	**4.922**	**+0.092**	**0.953**

Final 4.922 s · recheck 4.927 s · 6/6 < 5 s · 4.5 s target not reached

Reverse order + final recheck · 1 initialization + 1 warmup excluded · n = 3 per cohort

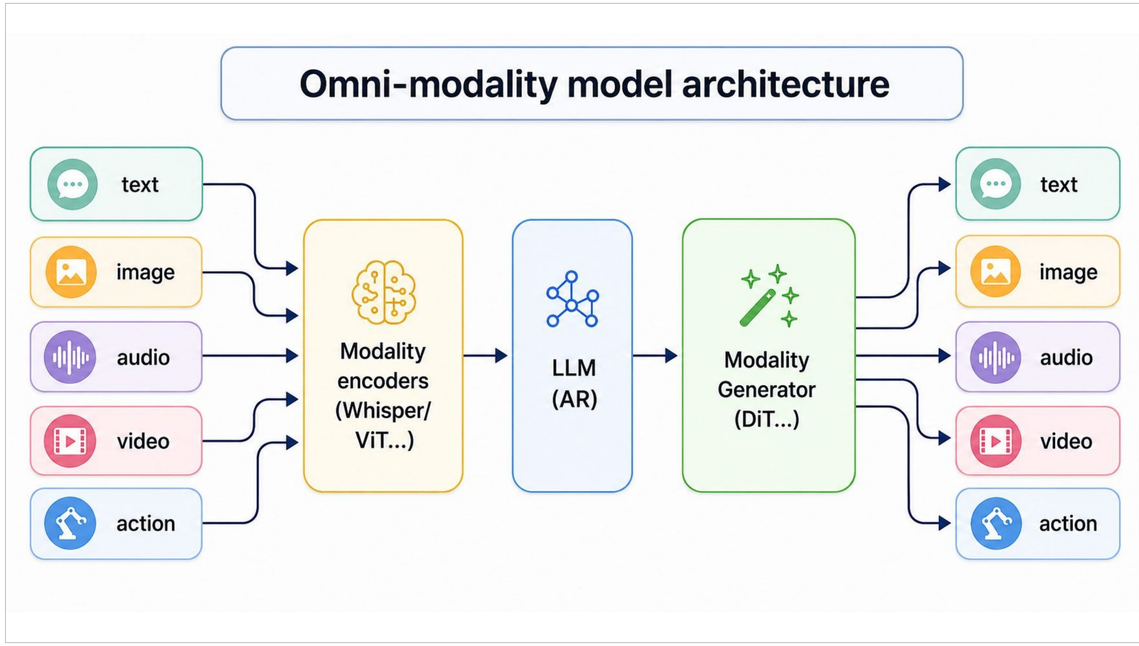
RTF = latency / 5.167 s. Model, schedule and precision changes are not all lossless. ASR / LLM are excluded.

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02 · ComfyUI Apps With vLLM-Omni

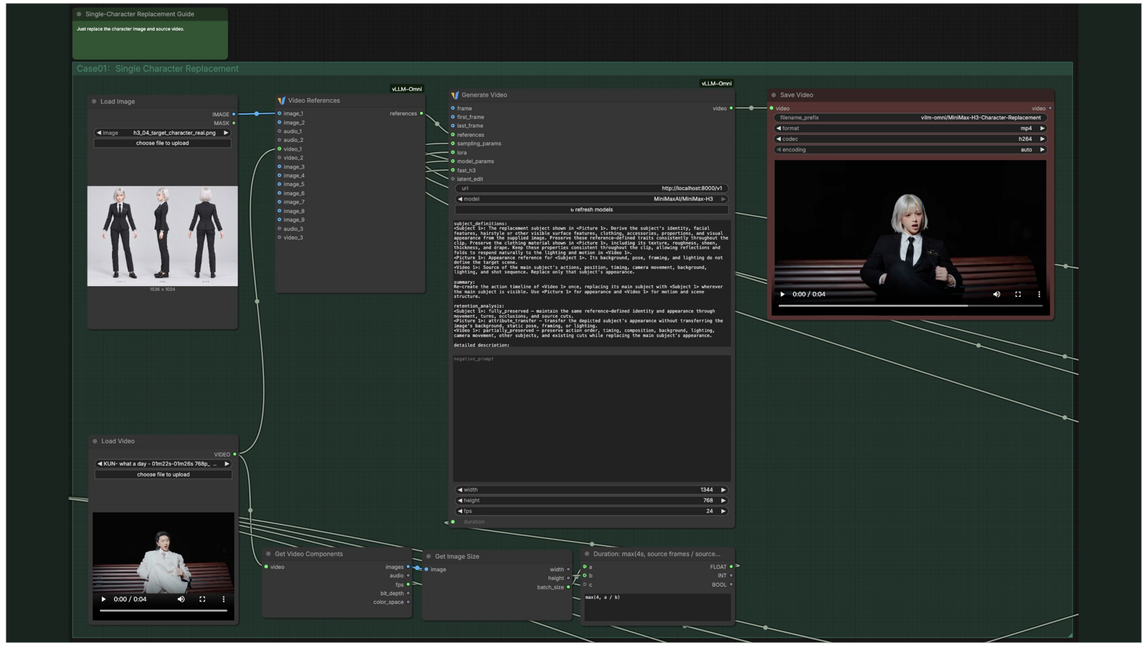
vLLM-Omni Brings Multi-GPU H3 to ComfyUI

PR #9194 adds character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI authors the workflow; vLLM-Omni runs scheduling, model execution, and multi-GPU deployment



ComfyUI Workflow · PR #9194

The workflow supports single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194

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Live Avatar: Configuration and Timing

Microphone → VAD / ASR → LLM → H3 audio + video → browser playback

OUTPUT

960 × 544 · 24 FPS

124-frame first window: 5.167 s

CONTINUATION

119 new frames: 4.958 s

5-frame overlap · video + audio history

HARDWARE

8 × SM120 GPUs

Recorded PRO 5000 node · BF16 RDMA

H3 generation

8 evaluations · QAD NVFP4 DiT · native FP4 VSA
TP1 / Ulysses8 · text TP8 / VAE PP8 · 2/3/2 heads

Decoder / history

36-layer NVFP4 VAE · MP4 pre-encoding
20% video-history noise · target video prefix off

Speech detection / ASR

CPU Silero VAD · 600 ms quiet endpoint
Qwen3-ASR-0.6B · BF16 · GPU 6, shared with H3

LLM / speech output

Qwen3-4B · BF16 · GPU 7, shared with H3
Streaming short replies · native H3 speech, no separate TTS

Two different clocks

Avatar generation

H3 request → complete MP4.
Excludes later audio screening and delivery.

Reply video starts

End of your speech → first reply frame.
Includes VAD, ASR, LLM, queueing, generation and playback wait.

One active call · fullscreen avatar · live captions · microphone + avatar audio recording

Demo shares GPUs 6 / 7 and carries history. Slide 18 is a standalone, fixed-input H3 benchmark with ASR / LLM stopped.

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